

Task 4: Mini Project Report

Molecular Docking Analysis of Aspirin with COX-2 Protein

1. Title

Molecular Docking Analysis of Aspirin with Cyclooxygenase-2 (COX-2) Protein

2. Abstract

Molecular docking is a computational technique used in drug discovery to predict the interaction between a drug molecule and a target protein. In this study, Aspirin was docked with the COX-2 protein using SwissDock. The docking results showed favorable binding interactions, indicating that Aspirin can bind to the active site of COX-2. This study demonstrates the application of bioinformatics tools in drug discovery.

3. Introduction

Cyclooxygenase-2 (COX-2) is an enzyme involved in inflammation and pain pathways. Aspirin is a widely used anti-inflammatory drug that inhibits cyclooxygenase enzymes. Molecular docking helps predict the binding orientation and affinity of a ligand toward a protein target.

4. Objectives

- To obtain the 3D structure of COX-2 protein.
- To obtain the structure of Aspirin.
- To perform molecular docking using SwissDock.
- To analyze the interaction between Aspirin and COX-2.

5. Materials and Methods

Protein:

COX-2 protein

PDB ID: 4LHF

Ligand:

Aspirin (Acetylsalicylic acid)

Databases:

Protein Data Bank (PDB)

PubChem

Docking Tool:

SwissDock

Procedure:

- Download COX-2 structure from PDB.
- Download Aspirin structure from PubChem.
- Upload protein and ligand to SwissDock.
- Run docking simulation.
- Analyze binding scores and interactions.

or WRR-99 (199) to cruzipain (1ewj), or or gabaremb (pub) to B-Rar (5nie).

1 - Submit a ligand

Provide a SMILES

... or upload a Mol2 file

... or input, or modify, or check the molecule **using the sketcher**

... or use the **advanced search**

✓

2 - Submit a target

Provide a PDB id (e.g. 5hie)

... or upload a PDB file 4LHF.pdb

... or use the **advanced search**

✓

2 - Define search space

3 - Define search space

Search box center Å

Search box size Å



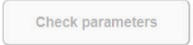
4 - Select parameters

Number of RIC 

Hide extra parameters

Sampling exhaustivity 

Cavity prioritization 

This job is estimated to take 0:02:07 (h:mm:ss)

5 - Start docking

Enter an email (optional)

Enter a docking name (optional)

START DOCKING [Reset form](#)

Navigation bar: [SwissDock](#) [SwissParam](#) [Swiss Sidechain](#) [SwissBioisostere](#) [SwissTargetPrediction](#) [SwissADME](#) [SwissSimilarity](#)

 **SwissDock**
SwissDrugDesign

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Parameters:

Submission date: June 10, 2026, 10:03 am UTC
Ligand: CC(=O)OC1=CC=CC=C1C(=O)O
Target: 4LHF.pdb
Box center: -79 - 6 - -2
Box size: 20 - 20 - 20
Docking method: AC
Sampling exhaustivity: medium
Cavity prioritization: medium
Number of RIC: 3
This job is estimated to take 0:02:15 (h:mm:ss)

Please, let this page open in your browser to be redirected to the result page automatically at the end of the calculations.

Bookmark [this link](#) to retrieve your results.

Calculation is in the queue. Number of jobs before yours: 0

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Query

Ligand CC(=O)OC1=CC=CC=C1C(=O)O

Target 4LHF.pdb

Method Attracting Cavities 2.0

Date June 10, 2026, 10:08 am UTC

Parameters:

Box center: -79 - 6 - -2 Sampling exhaustivity: medium Number of RIC: 3

Box size: 20 - 20 - 20 Cavity prioritization: medium

If you publish these results, please, cite the following papers:

Bugnon M, Röhrig UF, Goullieux M, Perez MAS, Daina A, Michielin O, Zoete V. SwissDock 2024: major enhancements for small-molecule docking with Attracting Cavities and AutoDock Vina. *Nucleic Acids Res* **2024**

Röhrig UF, Goullieux M, Bugnon M, Zoete V. Attracting Cavities 2.0: improving the flexibility and robustness for small-molecule docking. *J. Chem. Inf. Model.* **2023**

Ligand

Σ H O P O

Results

Export your results:

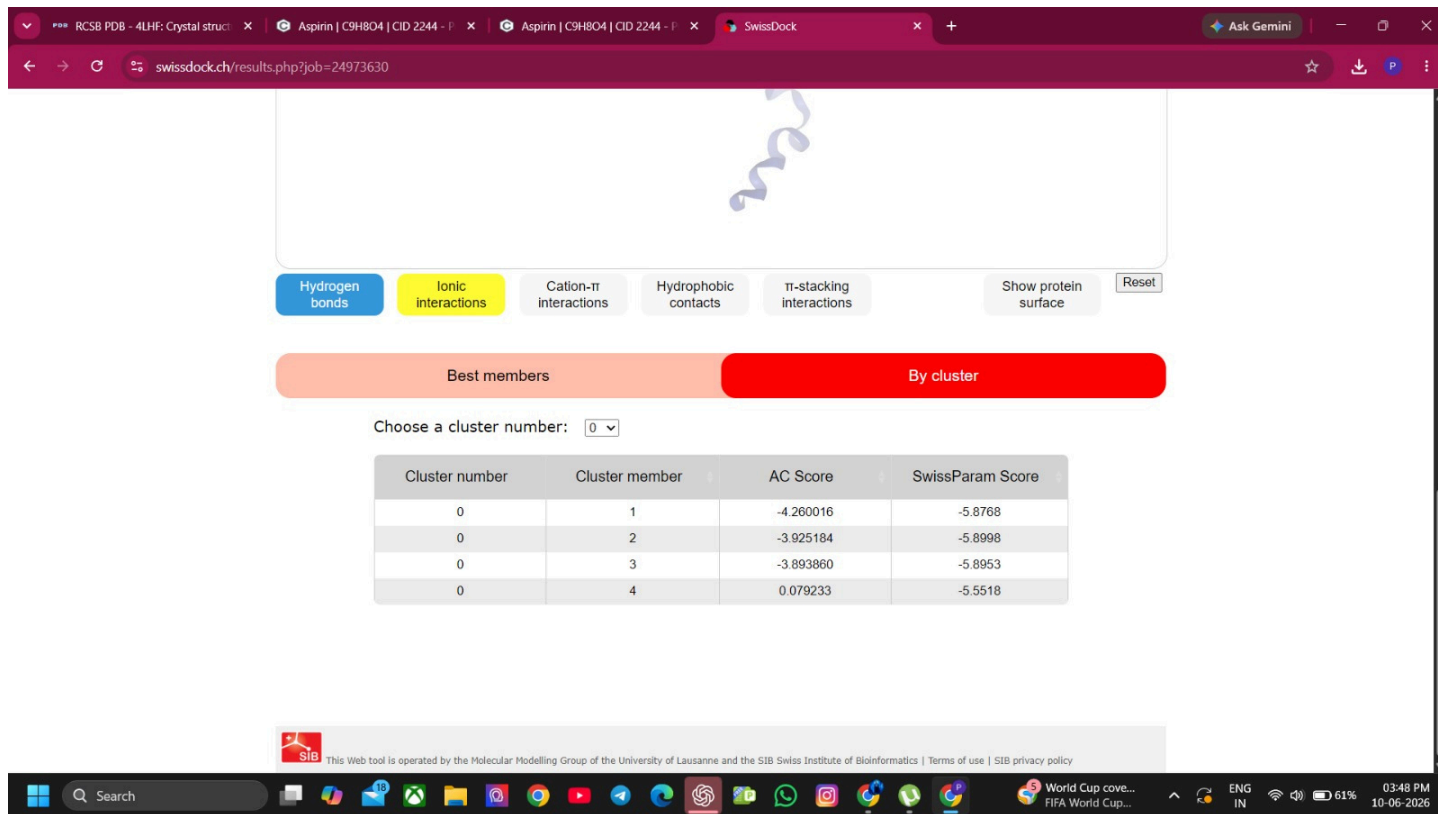
Hydrogen bonds **Ionic interactions** Cation-π interactions Hydrophobic contacts π-stacking interactions Show protein surface Reset

Best members By cluster

Number of displayed clusters: 10

Cluster number	Cluster member	AC Score	SwissParam Score
0	1	-4.260016	-5.8768
1	1	-3.067577	-5.8628
2	1	-2.677692	-5.7550
3	1	-2.205407	-5.7772
4	1	-1.828345	-5.7708
5	1	-1.823184	-5.6775
6	1	-1.268063	-5.6489
7	1	-0.974377	-5.6737
8	1	-0.970857	-5.5906
9	1	-0.670804	-5.5834

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6. Results

Cluster.	AC Score.	SwissParam Score
0.	-4.260	-5.877
1.	-3.068	-5.863
2.	-2.678	-5.755

The best docking pose was observed in Cluster 0, having the lowest AC score.

The negative binding score indicates favorable interaction between Aspirin and the COX-2 protein.

Conclusion

The study successfully demonstrated the docking of Aspirin with COX-2 using SwissDock. The docking score indicates a favorable interaction between the ligand and protein. This project highlights the importance of bioinformatics tools in drug discovery and molecular modeling.

Applications

- Drug discovery
- Structure-based drug design
- Identification of therapeutic targets
- Computational biology research

References

- Protein Data Bank (PDB)
- PubChem Database
- SwissDock Server
- Literature on COX-2 inhibitors